



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2022-2023
PREBOARD EXAMINATION

CLASS: X
SUBJECT: BIOLOGY [SET A]

FULL MARKS: 80
TIME: 2 HOURS

*Answers to this paper must be written on the paper provided separately.
You will not be allowed to write during the first 15 minutes.
This time is to be spent in reading the question paper.
The time given at the head of this paper is the time allowed for writing the answers.*

*Section A is compulsory. Attempt any four questions from Section B.
The intended marks for questions or parts of questions are given in brackets [].
This question paper consists of 6 printed pages.*

SECTION A (40 Marks)
(Attempt all questions from this section)

Question 1

Select the correct answers to the questions from the given options. (Do not copy the question. Write the correct answer only): [15]

- i. The solution which is used to identify the presence of starch:
a. Iodine, b. Alkaline Pyrogallous Acid, c. Soda Lime, d. Methylated spirit.
- ii. Guttation takes place through:
a. Stomach, b. Lenticels, c. Epidermis, d. Hydathodes.
- iii. Ovulation takes place on:
a. 8th -9th day, b. 13th -14th day, c. 19th -20th day, d. 26th - 27th day.
- iv. Which part of the eye is grafted in a needy patient from a donated eye?
a. Conjunctiva, b. Cornea, c. Choroid, d. Ciliary muscle.
- v. The number of deaths per 1000 of population per year:
a. Natality, b. Mortality, c. Population, d. Population density.
- vi. Fly ash and metallic ash are examples of:
a. Domestic waste, b. Biomedical waste, c. Agricultural waste, d. Industrial waste.

- vii. X rays contain a great deal of energy in the form of:
a. Electrons, b. Photons, c. Gamma, d. Neutron.
- viii. The pigment present in rod cells:
a. Iodopsin, b. Rhodopsin, c. Haemoglobin, d. None of them.
- ix. Sex chromosomes which determine the sex of an organism are also known as:
a. Autosomes, b. Allosomes, c. Squamous Epithelium, d. Cardiac muscle.
- x. Highly coiled, condensed chromatin fibre:
a. Chromosomes, b. Centrosome, c. Centromere, d. Chromomere.
- xi. The pigment present in urine:
a. Glucose, b. Bile, c. Urochrome, d. Protein
- xii. A sudden change in one or more genes or in the number or structure of chromosome:
a. Mutation, b. Karyotype, c. Karyokinesis, d. Cytokinesis
- xiii. The instrument which is used to measure the rate of transpiration:
a. Weighing machine, b. Bell Jar, c. Manometer, d. Potometer
- xiv. The number of water molecules required in the chemical reactions to produce one molecule of glucose during photosynthesis is:
a. 5, b. 12, c. 6, d. 7
- xv. The deficiency of ADH causes:
a. Diabetes mellitus, b. Diabetes insipidus, c. Dwarfism, d. Acromegaly

Question 2

- (i) Name the following: [5]
 - a. The hormone produced from adrenal medulla.
 - b. A condition in which some people by birth cannot discriminate between certain colours as red and green.
 - c. The growth movement of plant parts in response to touch stimulus.
 - d. An instrument used to measure blood pressure.
 - e. The chemical substance which initiates the process of clotting of blood.

- (ii) Choose the odd one out and write the correct category to which the others belong: [5]

Example- Large Intestine, Stomach, Mouth, Brain.

Odd one- Brain, Category- Organs of Digestive System

- a. Carbon Dioxide, Dirty dressings, Sulphur Dioxide, Carbon Monoxide.
- b. Condoms, Diaphragms, Pills, Sperm killing agents.

- c. Embryo, Morula, Foetus, Umbilical Cord .
 - d. Mineralocorticoids, Insulin, Glucocorticoids, Sex corticoids.
 - e. Utriculus, Sacculus, Semi-circular canal, Eustachian tube.
- (iii) Arrange and rewrite the terms in each group in the correct order so as to be in a logical sequence beginning with the term that is underlined. [5]
- a. Pinna, Oval Window, Cochlea, Auditory Canal.
 - b. Afferent Arteriole, Glomerulus, Proximal Tubule, Bowman's Capsule.
 - c. DNA, Chromatin Fibre, Chromosome, Nucleosomes .
 - d. Prophase, Telophase, Anaphase, Metaphase.
 - e. Aqueous Chamber, Retina, Vitreous Chamber, Lens.
- (iv) Match the correct answers of Column I with Column II. Please rewrite the correct pairs: [5]
- | | |
|--|---|
| <p>Column I</p> <ul style="list-style-type: none"> a. Lymphocyte b. Nucleosome c. Medulla Oblongata d. Leydig Cells e. Parthenocarpy | <p>Column II</p> <ul style="list-style-type: none"> a. Histone and DNA b. Antibodies c. Fruits without fertilization d. Phytohormone e. Progesterone f. Testosterone g. Breathing |
|--|---|
- (v) State the exact location of the following: [5]
- a. Incus
 - b. Ciliary body
 - c. Prostate gland
 - d. Atrio-ventricular node
 - e. Seminal vesicles

SECTION B (40 Marks)
(Attempt any four questions from this section)

Question 3

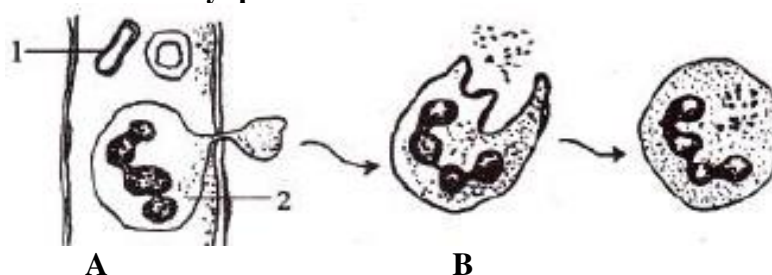
- (i) Define Phytohormones. [1]
- (ii) Mention the site of fertilization in a female human body. Mention the function of vas deferens. [2]
- (iii) Mention the changes observed in human eye during the light adaptation. [2]
- (iv) Identify the reflexes for the following: [2]
 - a. Coughing when the food swallowed enters the wind pipe
 - b. Applying brakes in your vehicle
- (v) Draw a neat labelled diagram to show the correction of Myopia. [3]

Question 4

- (i) Define Plasmolysis. [1]
- (ii) What do you understand by the term 'semi permeable membrane'? Mention one example of it. [2]
- (iii) Mention two important characteristic features of Metaphase. [2]
- (iv) A heterozygous tongue roller female (Rr) and a heterozygous tongue roller male (Rr) produces offsprings. Find out: [2]
 - a. Percentage of offsprings with homozygous non-tongue rolling characteristic
 - b. Percentage of offsprings with heterozygous tongue rolling characteristic
- (v) Draw a well labelled diagram of Anaphase with a cell having 4 chromosomes. [3]

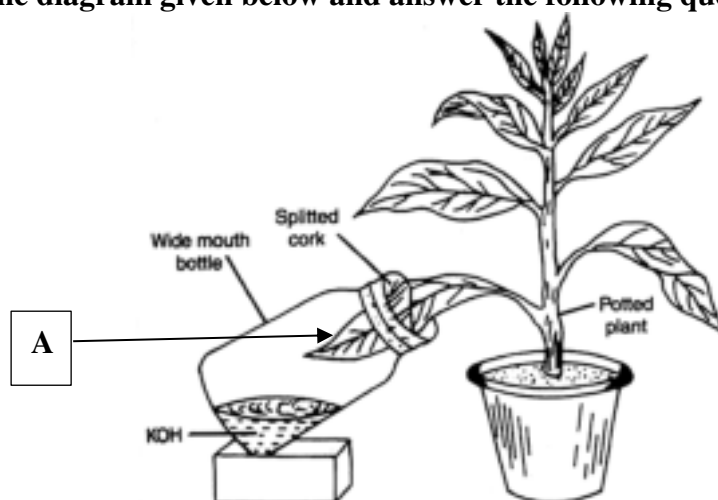
Question 5

- (i) Write the balanced chemical equation for photosynthesis. [1]
- (ii) Observe the diagram given below and define the process 'A' shown by the cell labelled as '2' and identify process 'B'. [2]



Vasectomy
Inverted red
triangle

- (iii) Mention the surgical method done in males for the control of population. Mention the sign for Family Welfare. [2]
- (iv) Mention two functions of spleen. [2]
- (v) Observe the diagram given below and answer the following questions: [3]



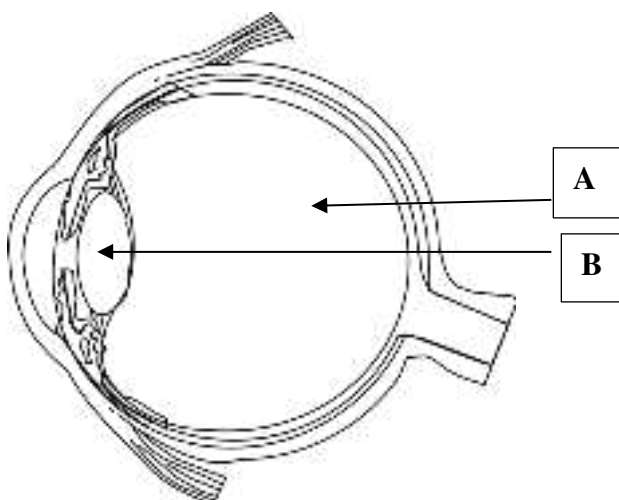
- a. Mention the aim of the experiment.
- b. What do you observe in leaf "A" at the end of the experiment when tested with Iodine solution?

Question 6

- (i) Mention the phenotypic ratio of F₂ generation in a dihybrid cross. [1]
- (ii) Mention two factors responsible for population explosion in India. [2]
- (iii) State the Law of Independent Assortment. Mention the scientific name for the plant used by Mendel in his experiment. [2]
- (iv) A pure plant with green pod (GG) is crossed with a pure plant with yellow pod (gg). Find out the Phenotypic ratio and Genotypic ratio of F₂ generation. [2]
- (v) Name the three bones present in the ear. [3]

Question 7

- (i) Name the paper used as an indicator of moisture in the experiment of transpiration. [1]
- (ii) Define Oogenesis. Name the female hormone which helps to maintain the uterine wall in a female body and is released by corpus luteum. [2]
- (iii) Differentiate between Mitosis and Meiosis (definition and where it occurs)[2]
- (iv) Observe the diagram of Human eye given below and label the parts marked as "A" and "B". [2]



- (v) Draw a neat, labelled diagram of a sperm. [3]

Question 8

- (i) Mention the full form of IAA. [1]
- (ii) Mention any two functions of Gibberellin. [2]
- (iii) Why do people in hilly regions take iodine in their diet? [2]
- (iv) Mention the two functions of Aqueous Humour. [2]

- (v) Observe the female reproductive system shown in the diagram given below and label 'A', 'B' and 'C'. [3]

